

### ■ A.1.3 Statistics

Here is part of a package that evaluates various simple statistical quantities for lists of numbers.

```
BeginPackage["Statistics`"]

Mean[list_List] := Apply[Plus, list] / Length[list]

GeometricMean[list_List] := Apply[Times, list]^(1/Length[list])

Median[list_List] :=
  Block[{s1,len},
    len = Length[list];
    s1 = Sort[list];
    If[OddQ[Length[s1]],
      s1[[len/2]],
      (s1[[len/2]] + s1[[len/2+1]])/2 ]
  ]

Variance[list_List] := Mean[ (list - Mean[list])^2 ]

`Range[list_List] := Apply[Max, list] - Apply[Min, list]
(* `Range hides System`Range *)

EndPackage[ ]
```

#### Simple statistics.

Read in the simple statistics package.

```
In[1]:= <<Statistics.m
```

Generate a list of ten pseudorandom numbers.

```
In[2]:= d = Table[Random[ ], {10}]
Out[2]= {0.353297, 0.0205034, 0.117374, 0.546051,
         0.417205, 0.704711, 0.559345, 0.247266, 0.133088,
         0.49174}
```

Here is the mean of the list of numbers.

```
In[3]:= Mean[d]
Out[3]= 0.359058
```

This evaluates the range of the list of numbers.

```
In[4]:= Range[d]
Out[4]= 0.684207
```

The function `Range` in the statistics package hides the built-in `Range` function. This is how you access the built-in `Range` function.

```
In[5]:= System`Range[4]
Out[5]= {1, 2, 3, 4}
```

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